ModelIt!

Student Activity Pack

Why Hot Cheetos Make You Cry

How Your Body's Alarm System Detects and Responds to the World

Grade 7 | NGSS MS-LS1-8 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 How does your body know when something is hot, cold, or painful?

2 Why do you think spicy food makes some people cry?

3 Draw what you think happens inside your body between touching a hot pan and pulling your hand away.

4 Can your senses be tricked? Give an example.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Stimulus Intensity | Receptor Sensitivity | Nerve Signal Strength | Brain Response

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Mild Stimulus

Low stimulus intensity, normal receptor sensitivity

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Hot Cheetos Attack

Maximum stimulus intensity (capsaicin binding to pain receptors)

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Sensory Adaptation

Sustained stimulus over time

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Reflex Arc Speed — Some signals bypass the brain entirely — reflexes go from receptor to spinal cord to muscle for faster response

Sensory Adaptation — Receptors decrease their firing rate when exposed to a constant stimulus over time

Pain Threshold — The minimum stimulus intensity required to trigger a pain response — varies between individuals

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does your body know the difference between hot, cold, sweet, and pain — and why do Hot Cheetos trick your brain into thinking your mouth is on fire? Use evidence from your simulation to support your answers.

1 Why does your hand pull away from a hot stove BEFORE you consciously feel the pain?

2 If capsaicin doesn't actually damage your mouth, why does your brain respond as if it does?

3 Why do you eventually stop noticing a strong smell after being around it for a while?

4 How would life be different if you had no pain receptors?

5 How does understanding sensory receptors help doctors treat conditions like chronic pain?

8 STEM Challenge: Design a Sensory Experiment

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YOUR ENGINEERING MISSION

Design and conduct a controlled experiment testing how different variables affect sensory perception — then explain your results using your nervous system model.

THE SCENARIO

A neuroscience lab needs student researchers to investigate sensory perception. Your team will design an experiment, collect data, and present findings about how the nervous system processes sensory information.

Engineering Design Thinking

1. WHAT SPECIFIC SENSORY QUESTION ARE YOU TESTING (TASTE, TOUCH, TEMPERATURE, SIGHT)?

2. WHAT IS YOUR INDEPENDENT VARIABLE AND HOW WILL YOU CHANGE IT?

3. WHAT IS YOUR DEPENDENT VARIABLE AND HOW WILL YOU MEASURE IT?

4. HOW MANY TRIALS WILL YOU RUN AND WHY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Neuroscientists study the nervous system to understand how we perceive the world, form memories, and respond to stimuli. They work in hospitals, research labs, and pharmaceutical companies, earning $70,000–$120,000/year.